Representations of Integral Quadratic Forms by Sums of Squares

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Abstract

Let $g[n]$ be the minimum number of squares whose sum represents all positive definite integral quadratic forms of rank n which are represented by sums of squares. In this article, we first discuss representations of integers by unimodular lattices. We then estimate the order of magnitude of the diameter of the 2-graph of unimodular lattices of rank n. Combining these results we prove $g[n] = O(3^{n/2}\, n \log n)$. We also provide a lower bound for $g[n]$. Finally, we discuss s-integrable lattices as an application of our method.

1 Introduction

In 1770, Lagrange [L] proved the famous four square theorem, i.e., for every positive integer n, there exists an integer solution of the equation

$$x^2 + y^2 + z^2 + w^2 = n.$$

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After Lagrange, this theorem was generalized in many directions. One interesting generalization is so called the *new* (or *quadratic*) *Waring's Problem* due to L. J. Mordell [M1], which is about sums of squares that represent all positive integral quadratic forms of given rank. In particular, Mordell proved that every binary positive integral quadratic form can be represented by a sum of five squares. Later, C. Ko [K1] proved that every ternary (quaternary or quinary) positive integral quadratic form can be represented by a sum of six (seven or eight, respectively) squares. So, they naturally expected that every positive n-ary integral quadratic form can be represented by a sum of $(n+3)$ squares. This, however, turned out to be false. The quadratic form defined by the root lattice E_6 cannot be represented by sums of squares (see [M2], [CS2] and [Pl]). After Mordell found this, they tried to determine the minimum number $g[n]$ of squares whose sum represents all positive integral quadratic forms of rank n that are represented by sums of squares.

We adopt lattice theoretic language. A $\mathbb{Z}$-*lattice* L is a finitely generated free $\mathbb{Z}$-module in $\mathbb{R}^n$ equipped with a non-degenerate symmetric bilinear form B such that $B(L,L) \subseteq \mathbb{Z}$. The corresponding quadratic map is denoted by Q, i.e., $Q(\mathbf{e}) = B(\mathbf{e},\mathbf{e})$ for every $\mathbf{e} \in L$. The ideal of $\mathbb{Z}$ generated by $B(\mathbf{e}_i, \mathbf{e}_j)$'s is called the *scale* of L, denoted by $\mathfrak{s}(L)$. For a $\mathbb{Z}$-lattice

$$L = \mathbb{Z}\mathbf{e}_1 + \mathbb{Z}\mathbf{e}_2 + \cdots + \mathbb{Z}\mathbf{e}_m$$

with basis $\{\mathbf{e}_1, \mathbf{e}_2, \cdots, \mathbf{e}_m\}$, we define its corresponding matrix

$$M_L := (B(\mathbf{e}_i, \mathbf{e}_j)),$$

which is an $m \times m$ symmetric integer matrix. We often identify a $\mathbb{Z}$-*lattice* L with its corresponding matrix M_L. If M_L is diagonal, we often write

$$L = \langle Q(\mathbf{e}_1), Q(\mathbf{e}_2), \cdots, Q(\mathbf{e}_n)\rangle \quad \text{or} \quad \langle Q(\mathbf{e}_1)\rangle \perp \langle Q(\mathbf{e}_2)\rangle \perp \cdots \perp \langle Q(\mathbf{e}_n)\rangle.$$

For $a \in \mathbb{R}^\times$, we let

$$aL := \mathbb{Z}(a\mathbf{e}_1) + \mathbb{Z}(a\mathbf{e}_2) + \cdots + \mathbb{Z}(a\mathbf{e}_n).$$

The lattice $\mathbb{Z}^n$ equipped with the standard inner product is denoted by I_n. A $\mathbb{Z}$-lattice L is said to be *positive definite* or simply *positive* if $Q(\mathbf{e}) > 0$ for every $\mathbf{e} \in L, \mathbf{e} \neq \mathbf{0}$. In this paper, we always assume the following unless stated otherwise:

Every $\mathbb{Z}$-lattice is positive definite.

A $\mathbb{Z}$-*lattice* L is said to be *even* if $Q(L) \subseteq 2\mathbb{Z}$, and *odd*, otherwise. As usual,

$$dL := \det(B(\mathbf{e}_i, \mathbf{e}_j))$$

is called the *discriminant* of L. L is said to be *unimodular* if $dL = 1$. For a $\mathbb{Z}$-lattice L and a prime p,

$$L_p := \mathbb{Z}_p L = \mathbb{Z}_p \otimes_{\mathbb{Z}} L$$

is a $\mathbb{Z}_p$-lattice called the *localization* of L at p, where $\mathbb{Z}_p$ is the p-adic integer ring.

Let ℓ, L be $\mathbb{Z}$-lattices. We say that L *represents* ℓ if there is an injective $\mathbb{Z}$-linear map from ℓ into L that preserves the bilinear forms, and write $\ell \to L$. Such a map is called a *representation*. A representation is called an *isometry* if it is surjective. We say that ℓ and L are *isometric* and write $\ell \cong L$ if there exists an isometry between them. For a $\mathbb{Z}$-lattice L, we define the *class* and *genus* of L by

$$\begin{aligned} \mathrm{cls}(L) &:= \{\, K : \mathbb{Z}\text{-lattice} \,|\, K \simeq L \text{ over } \mathbb{Z} \,\}, \\ \mathrm{gen}(L) &:= \{\, K : \mathbb{Z}\text{-lattice} \,|\, K_p \simeq L_p \text{ over } \mathbb{Z}_p \text{ for all } p \,\}. \end{aligned}$$

It is well known that $\mathrm{gen}(L)$ contains a finite number of distinct classes. This number is called the *class number* of L, denoted by $h(L)$. If $\ell_p \to L_p$ for all p, or equivalently $\ell \to \mathrm{gen}(L)$, then it is known that $\ell \to K$ for some $K \in \mathrm{gen}(L)$.

Let

$$\mathfrak{S}_n := \{\, \ell : \mathbb{Z}\text{-lattices} \mid \ell \to I_g \text{ for some } g,\ \mathrm{rank}(\ell) = n \,\},$$

and

$$g[n] := \min\{\, g \mid \ell \to I_g \text{ for every } \ell \in \mathfrak{S}_n \,\}.$$

Applying the results in [HKK], one can easily prove that $g[n]$ exists for all n. The four square theorem of Lagrange, and the results of Mordell's and Ko's mentioned above can be summarized as the following:

$$g[n] = n + 3 \quad \text{for } 1 \le n \le 5.$$

In fact, this is an immediate consequence of local representation theory and the fact that the class number of I_n is one for $1 \leq n \leq 8$. The following question arises quite naturally:

$$g[n] = n + 3 \ \ \textit{for all} \ \ n \geq 1 \ \textit{?}$$

Concerning this question, Ko conjectured in [K2] that $g[6] = 9$. However, the authors proved recently [KO1][KO2] that

$$g[n] \geq \lfloor 3n/2 \rfloor + 1 > n + 3 \ \text{ for all } n \geq 6, \ \text{ and } \ g[6] = 10.$$

The first explicit upper bound for $g[n]$ was given by Icaza [Ic]. She obtained her bound by computing the so called HKK-constant (see [HKK]). But her bound is too big containing a factor $n^{n+1}2^{4h(I_{n+3})}$, where $h(I_n)$ denotes the class number of the genus of I_n which can be shown to be of the form $h(I_n) = n^{\Theta(n^2)}$ for some arithmetic function $\Theta(n^2)$ bounded between constant multiples of n^2 for all $n >> 0$.

In section 2 of this paper, we prove that every odd (even) unimodular $\mathbb{Z}$-lattice of rank bigger than three represents all (even, respectively) integers greater than or equal to $(n/2)^4$. In section 3, we compute a lower bound and an upper bound of the diameter of the 2-graph of unimodular lattices of rank n. And then from these results, we provide a sharper upper bound, of size $O(3^{n/2}\, n \log n)$, for $g[n]$ in section 4. Furthermore, we give a lower bound of $g[n]$ and $\phi(s)$ (for the definition of $\phi(s)$, see section 4).

For unexplained terminology, notation, and basic facts about $\mathbb{Z}$-lattices, we refer the readers to O'Meara [O'M1], Conway and Sloane [CS1].

2 Representation of integers by unimodular $\mathbb{Z}$-lattices

Let L be a unimodular $\mathbb{Z}$-lattice of rank n and $\mu(L)$ be the smallest positive integer that is represented by L. It was proved in [MOS] for even case and in [RS] for odd case that

$$\mu(L) \leq 2\lfloor n/24 \rfloor + 2$$

unless L is odd unimodular of rank 23. In the exceptional case, $\mu(L) \leq 3$ (see [RS]). In contrast with these results, Conway and Thompson proved

(unpublished, see [MH]) that there exists a unimodular $\mathbb{Z}$-lattice L of rank n with $\mu(L) \geq k(n)$, where $k(n)$ denotes the closest integer to $(\frac{5}{3\omega_n})^{2/n}$. Here

$$\omega_n = \pi^{n/2} \left(\Gamma((n+2)/2)\right)^{-1}$$

is the volume of the n-dimensional unit ball in $\mathbb{R}^n$ and therefore $k(n)$ is asymptotically $\frac{n}{2\pi e}$. For a $\mathbb{Z}$-lattice K, we define

$$c(K) := \min\{\, m \mid \langle k \rangle \to K \ \text{ for all (even, if } K \text{ is even) integers } k \geq m \,\}.$$

Let

$$\begin{aligned} c_{\mathrm{e}}(n) &:= \max\{\, c(K) \mid K : \text{ even unimodular of rank } n \,\}, \\ c_{\mathrm{o}}(n) &:= \max\{\, c(K) \mid K : \text{ odd unimodular of rank } n \,\}, \end{aligned}$$

and

$$c(n) := \max\{\, c_{\mathrm{e}}(n), c_{\mathrm{o}}(n) \,\}.$$

If $n \geq 4$, then $c(n)$ always exists (see [T]). For the exact values and upper bounds of $c_{\mathrm{e}}(n)$ for lower rank even unimodular $\mathbb{Z}$-lattices, see [CLR], [OS], [LR] and [Pe]. In this section, we give an upper bound for $c(n)$ for $n \geq 4$. We begin with the following trivial but useful lemma.

Lemma 2.1 *Let K be any $\mathbb{Z}$-lattice of any rank n with $\mu(K) = \mu$. For a positive integer $t \geq \mu$, let $r(t, K)$ be the number of vectors $\mathbf{x} \in K$ such that $Q(\mathbf{x}) = t$. Then*

$$r(t, K) \leq \left(2\sqrt{t/\mu} + 1\right)^n - \left(2\sqrt{t/\mu} - 1\right)^n.$$

Proof. We may assume that $K \subset \mathbb{R}^n$. Let $V_n(r) = \omega_n r^n$ be the volume of the n-dimensional ball of radius r in $\mathbb{R}^n$. Let $X := \{\, \mathbf{x} \in K \mid Q(\mathbf{x}) = t\}$. Then the balls with radius $\sqrt{\mu}/2$ centered at the points in X do not intersect each other. Therefore

$$r(t, K) \cdot V_n\left(\sqrt{\mu/4}\right) \leq V_n\left(\sqrt{t} + \sqrt{\mu/4}\right) - V_n\left(\sqrt{t} - \sqrt{\mu/4}\right).$$

The lemma follows immediately. $\square$

Remark. We call $r(t, K)$ the representation number of t by K. Clearly $r(1, K) \leq 2n$. For $n \geq 16$, one can easily show that $r(2, K) \leq 2n(n-1)$ by considering all root lattices. For $n \geq 5$, the asymptotic behavior for $r(t, K)$ is well known (see [H], [P] and [I]).

Theorem 2.1 *For $n \geq 4$,*

$$c(n) \leq (n/2)^4 .$$

Proof. Let L be a unimodular $\mathbb{Z}$-lattice of rank n. Since all unimodular $\mathbb{Z}$-lattices of rank less than or equal to 24 are classified (see, for example, [CS1]), one can easily check the above assertion for those $\mathbb{Z}$-lattices. Therefore, for computational convenience, we may assume that

$$n \geq 20 \qquad \text{and} \qquad \mu(L) \geq 2. \tag{1}$$

We follow Iwaniec [I, chapter 11], where he introduced the circle method for the representation number of an integer by a quadratic form of rank ≥ 4. For details, we refer the readers to his book. To prove the theorem, we must compute all the error terms quite accurately. Let $k = n/2$ and $C \geq 1$ be any real number. As usual, $e(z) = e^{2\pi iz}$. We define the adjoint quadratic form

$$Q^* := (1/2)A^{-1}[\mathbf{x}]$$

of Q, where $A = 2M_L$ and $A[\mathbf{x}] := {}^t\mathbf{x}A\mathbf{x}$.

If we define

$$T(c,t;x) = |A|^{-1/2}c^{-n}\left(\frac{i}{z}\right)^k \sum_{\mathbf{m}\in\mathbb{Z}^n} T_{\mathbf{m}}(c,t;x)e\left(\frac{-Q^*(\mathbf{m})}{c^2 z}\right), \tag{2}$$

where c is a positive integer less than or equal to C and

$$T_{\mathbf{m}}(c,t;x) = \sum_{\substack{C<d\leq c+C \\ gcd(c,d)=1,\, cdx<1}} e\left(\frac{t\,\overline{d}}{c}\right) \sum_{\mathbf{h}\,(\mathrm{mod}\ c)} e\left(-\frac{d}{c}(Q(\mathbf{h}) + {}^t\mathbf{h}\mathbf{m})\right), \tag{3}$$

then

$$r(t,L) = 2\ \mathrm{Re} \sum_{c=1}^{\lfloor C \rfloor} \int_0^{\frac{1}{cC}} T(c,t;x)e(-tz)dx. \tag{4}$$

Here, $z = x + iy \in \mathbb{C}$ with $y > 0$, $\mathrm{Re}\, z = x$, and $\overline{d}$ is the multiplicative inverse of $d \pmod c$. If $0 < x < \frac{1}{c(c+C)}$, then $T_{\mathbf{m}}(c,t;x)$ is independent of x.

In that case, we simply write $T_{\mathbf{m}}(c,t)$. Now we divide (4) into the following five parts:

$$\begin{aligned}
r(t,L) &= |A|^{-1/2}\sum_{c=1}^{\infty} c^{-n}T_{\mathbf{0}}(c,t)\int_{-\infty}^{\infty}\left(\frac{i}{z}\right)^k e(-tz)dx && \cdots(r_1)\\
&-|A|^{-1/2}\sum_{c=\lfloor C\rfloor+1}^{\infty} c^{-n}T_{\mathbf{0}}(c,t)\int_{-\infty}^{\infty}\left(\frac{i}{z}\right)^k e(-tz)dx && \cdots(r_2)\\
&+2\ \mathrm{Re}\ \sum_{c=1}^{\lfloor C\rfloor}|A|^{-1/2}c^{-n}\int_{1/c(c+C)}^{1/cC} T_{\mathbf{0}}(c,t;x)\left(\frac{i}{z}\right)^k e(-tz)dx && \cdots(r_3)\\
&+2\ \mathrm{Re}\ \sum_{c=1}^{\lfloor C\rfloor}|A|^{-1/2}c^{-n}\int_{0}^{1/cC}\left(\frac{i}{z}\right)^k \times\\
&\qquad\sum_{\mathbf{m}\in\mathbb{Z}^n\setminus\{\mathbf{0}\}} T_{\mathbf{m}}(c,t;x)e\left(\frac{-Q^*(\mathbf{m})}{c^2z}\right)e(-tz)dx && \cdots(r_4)\\
&-2\ \mathrm{Re}\ \sum_{c=1}^{\lfloor C\rfloor}|A|^{-1/2}c^{-n}T_{\mathbf{0}}(c,t)\int_{1/c(c+C)}^{\infty}\left(\frac{i}{z}\right)^k e(-tz)dx\,. && \cdots(r_5)
\end{aligned}$$

First assume that L is odd unimodular. Let

$$G_{c,d}(Q,\mathbf{m}) := \sum_{\mathbf{h}\,(mod\ c)} e\left(-\frac{d}{c}(Q(\mathbf{h})+{}^t\mathbf{hm})\right) \tag{5}$$

be the Gauss sum factor in (3). By a suitable transformation, we may assume that $Q(\mathbf{h})\equiv h_1^2+h_2^2+\cdots+h_n^2 \pmod c$. Then

$$G_{c,d}(Q,\mathbf{m}) = \prod_{i=1}^{n}\sum_{h\,(mod\ c)} e\left(-\frac{d}{c}(h^2+m_ih)\right),$$

where $\mathbf{m}=(m_1,m_2,\cdots,m_n)$. Therefore by [S, Lemma 1], we obtain

$$|G_{c,d}(Q,\mathbf{m})| \le (gc)^k, \tag{6}$$

where $g=gcd(2,c)$. From this and (3) follows

$$|T_{\mathbf{m}}(c,t;x)| \le 2^{k-1}c^{k+1} \ \text{ for all } \mathbf{m},\ x \text{ and } t. \tag{7}$$

Now we take $C=\sqrt{t}$ and $y=1/t$. Then

$$\int_{-\infty}^{\infty}\left(\frac{i}{z}\right)^k e(-tz)dx = (2\pi)^k\Gamma(k)^{-1}t^{k-1},$$

and

$$\sum_{c=1}^{\infty} c^{-n} T_{\mathbf{0}}(c,t) = \prod_{p:\, prime} \alpha_p(t,L),$$

where $\alpha_p(t,L)$ is the local density at p. Therefore

$$r_1 = \pi^k \Gamma(k)^{-1} t^{k-1} \prod_{p:prime} \alpha_p(t,L).$$

Now we compute the upper bounds of the absolute values for the other r_i's by using Lemma 2.1 and (7). One can easily show the following:

$$\begin{aligned}
|r_2| &\leq \frac{2^{k-1}\pi^k t^{k-1}}{\Gamma(k)} \sum_{c=\lfloor\sqrt{t}\rfloor+1}^{\infty} c^{1-k} \leq \frac{2^{2k-1}\pi^k t^{k/2}}{\Gamma(k)}, \\
|r_3| &\leq \sum_{c=1}^{\lfloor\sqrt{t}\rfloor} \frac{e^{2\pi}}{k-1} \left((c+\sqrt{t})^{k-1} - \sqrt{t}^{\,k-1} \right) \leq \frac{2^{k-1}e^{2\pi}}{k-1} t^{k/2}, \\
|r_5| &\leq \frac{2^{k-1}e^{2\pi}}{k-1} t^{k/2}.
\end{aligned}$$

Since L is unimodular,

$$\begin{aligned}
\left| \sum_{\mathbf{m}\in\mathbb{Z}^n\setminus\{\mathbf{0}\}} e\left(\frac{-Q^*(\mathbf{m})}{c^2 z} \right) \right| &\leq \sum_{\mathbf{m}\in\mathbb{Z}^n\setminus\{\mathbf{0}\}} e\left(\frac{-\pi y}{2c^2(x^2+y^2)} Q^{-1}[\mathbf{m}] \right) \\
&= \sum_{m=1}^{\infty} r(m,L) e\left(\frac{-\pi y m}{2c^2(x^2+y^2)} \right).
\end{aligned}$$

Let $\beta = \dfrac{\pi y}{2c^2(x^2+y^2)}$. We may assume that $r(m,L) \leq (4m)^k$ by the assumption (1). Therefore

$$\sum_{m=1}^{\infty} r(m,L) e^{-\beta m} \leq 4^k \int_0^{\infty} x^k e^{-\beta x} dx \leq \frac{4^k \Gamma(k+1)}{\beta^{k+1}}. \tag{8}$$

This implies

$$\begin{aligned}
|r_4| &\leq \frac{e^{2\pi} 2^{3k+1} \Gamma(k+1) t^{k+1}}{\pi^{k+1}} \sum_{c=1}^{\lfloor\sqrt{t}\rfloor} c^{k+3} \int_0^{1/c\sqrt{t}} |z|^{k+2} dx \\
&\leq \frac{e^{2\pi} 2^{(7k+4)/2} \Gamma(k+1)}{\pi^{k+1}} t^{k/2}.
\end{aligned}$$

By Lemma 9.4 of [MH],

$$\frac{5}{6} \le \prod_p \alpha_p(t, L) \le \frac{6}{5}.$$

Therefore

$$r(t, L) \ge \left(\frac{5\pi^k}{6\Gamma(k)}\right) t^{k-1} - \left(\frac{2^{2k-1}\pi^k}{\Gamma(k)} + \frac{2^k e^{2\pi}}{k-1} + \frac{e^{2\pi} 2^{(7k+4)/2}\Gamma(k+1)}{\pi^{k+1}}\right) t^{k/2}.$$

Consequently, if $t \ge k^4$, then $r(t, L) > 0$, which proves the theorem for odd unimodular case.

Now assume that L is even. In this case, we may assume that $Q(\mathbf{h}) \equiv 2h_1h_2 + \cdots + 2h_{n-1}h_n \pmod{c}$ by a suitable transformation. Then, one can easily show that

$$|T_{\mathbf{m}}(c, t; x)| \le 2^{k-1} c^{k+1},$$

and

$$\frac{25}{16} \le \prod_p \alpha_p(2t, L) \le \frac{18}{7},$$

just by comparing the 2-adic densities of even and odd cases. This completes the proof. $\square$

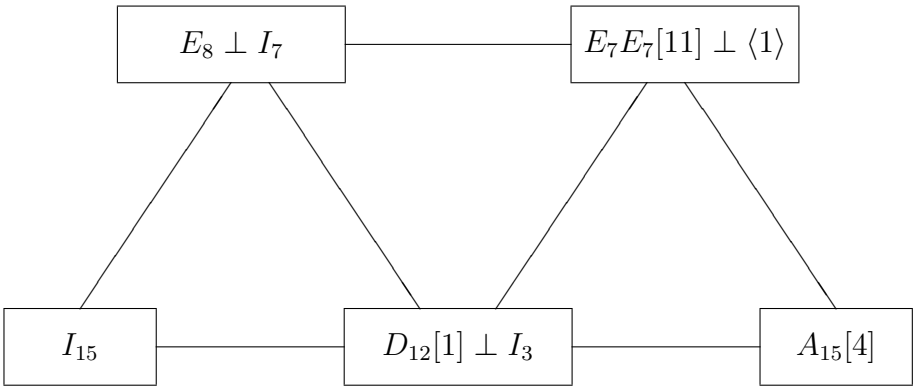
3 The 2-graph of unimodular $\mathbb{Z}$-lattices

Let $\mathfrak{U}_n$ be the set of all unimodular (both even and odd) $\mathbb{Z}$-lattices of rank n. Two $\mathbb{Z}$-lattices $K, L \in \mathfrak{U}_n$ are said to be 2-neighbors if there exists a $\mathbb{Z}$-lattice L' isometric to L such that

$$[L' : L' \cap K] = [K : L' \cap K] = 2.$$

In order to define the *2-graph* $G(\mathfrak{U}_n : 2)$ of $\mathfrak{U}_n$, we take for the vertices the isometric classes of unimodular lattices in $\mathfrak{U}_n$. Two vertices are connected by a simple edge when and only when they are 2-neighbors. This definition is due to Kneser, which is slightly different from those of [BH] and [CS1]. It is well known that $G(\mathfrak{U}_n : 2)$ is connected (see [Kn] or [O'M1; 106.4]). The following figure illustrates the 2-graph $G(\mathfrak{U}_{15} : 2)$ of $\mathfrak{U}_{15}$.

Figure 3.1.



For the gluing notations for $\mathbb{Z}$-lattices and the 2-graph of even unimodular lattices of rank 24, see [CS1], [CS2]. By abuse of notations, we will regard individual lattices in $\mathfrak{U}_n$ as representatives of their classes which are vertices of the graph.

For a vertex $L \in \mathfrak{U}_n$, let $\Delta(L)$ denote the number of vertices adjacent to L. For two vertices $L, K \in \mathfrak{U}_n$, define the *distance* between L and K by the minimum number of edges between L and K and write $d(L, K)$. Let

$$\mathfrak{d}(L) := max\{\, d(L, K) \,|\, K \in \mathfrak{U}_n \,\} \text{ and } \mathfrak{d}(n) := max\{\, \mathfrak{d}(L) \,|\, L \in \mathfrak{U}_n \,\}.$$

We call $\mathfrak{d}(n)$ the *diameter* of $G(\mathfrak{U}_n : 2)$. It is well known [Kn] that every unimodular $\mathbb{Z}$-lattice that is adjacent to I_n is of the form $I_{n-4k} \perp D_{4k}[1]$. Thus we obtain

$$\Delta(I_n) \leq \lfloor n/4 \rfloor.$$

It is also well known [Kn] that

$$\mathfrak{d}(n) \leq 2 \quad \text{for all} \quad n \leq 16. \tag{9}$$

Lemma 3.1 *For $L \in \mathfrak{U}_n$,*

$$\Delta(L) \leq \begin{cases} 2\left(\begin{pmatrix} n \\ 4 \end{pmatrix} + \begin{pmatrix} n \\ 8 \end{pmatrix} + \cdots + \begin{pmatrix} n \\ 4\lfloor n/4 \rfloor \end{pmatrix}\right) & \text{if } L \text{ is odd,} \\ 2^n + 2^{n/2} - 2 & \text{if } L \text{ is even.} \end{cases}$$

Proof. Let $L = \mathbb{Z}\mathbf{x}_1 + \mathbb{Z}\mathbf{x}_2 + \cdots + \mathbb{Z}\mathbf{x}_n$ be any unimodular $\mathbb{Z}$-lattice of rank n. First assume that L is odd. By the weak approximation theorem, we may also assume that $B(\mathbf{x}_i, \mathbf{x}_j) \equiv \delta_{ij} \pmod 8$. Any unimodular lattice K adjacent to L is isometric to

$$L_{\mathbf{y}} = \mathbb{Z}\mathbf{y} + \{\mathbf{x} \in L \mid B(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}\}$$

for some $\mathbf{y} \in (1/2)L - L$ and $Q(\mathbf{y}) \in \mathbb{Z}$ (see [O'M1]). Note that

$$L_{\mathbf{y}} = L_{\mathbf{y}'} \quad \text{if and only if} \quad \mathbf{y} - \mathbf{y}' \in L \ \text{ and } \ B(\mathbf{y}, \mathbf{y}') \in \mathbb{Z}.$$

Define

$$S := \{\mathbf{a} = (a_1, a_2, \cdots, a_n) \mid a_i \in \{0, 1, 2, 3\},\ \mathbf{a} \notin 2\mathbb{Z}^n,\ \sum a_i^2 \equiv 0 \pmod 4\},$$

and give an equivalence relation on S by

$$\mathbf{a} \sim \mathbf{b} \quad \text{if and only if} \quad a_i \equiv b_i \pmod 2 \ \text{ and } \ \sum a_i b_i \equiv 0 \pmod 4.$$

If we define

$$\mathbf{y}(\mathbf{a}) := (1/2)(a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_n\mathbf{x}_n)$$

for $\mathbf{a} = (a_1, a_2, \cdots, a_n) \in S$, then $L_{\mathbf{y}(\mathbf{a})}$ is adjacent to L and every $L_{\mathbf{y}}$ adjacent to L can be obtained in this way. Furthermore, $L_{\mathbf{y}(\mathbf{a})} = L_{\mathbf{y}(\mathbf{b})}$ if and only if $\mathbf{a} \sim \mathbf{b}$. Note that they might be isometric to each other. Thus we obtain

$$\Delta(L) \le |S/\sim|$$

and hence the desired inequality follows.

Now assume that L is even. Then $n \equiv 0 \pmod 8$. We may assume that

$$M_L \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \perp \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \perp \cdots \perp \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \pmod 8.$$

We omit the rest, which is similar to the odd unimodular case. □

Lemma 3.2 *For $n \ge 17$,*

$$\mathfrak{d}(I_n) \le \sum_{i=17}^{n} \lfloor 2 \log_2 i \rfloor - n + 18.$$

Proof. Let $L = \mathbb{Z}\mathbf{x}_1 + \mathbb{Z}\mathbf{x}_2 + \cdots + \mathbb{Z}\mathbf{x}_n \in \mathfrak{U}_n$ for $n \geq 17$. Since

$$4^{\lfloor 2\log_2 n\rfloor - 1} \geq (n/2)^4 \geq c(n),$$

L should represent $4^{\lfloor 2\log_2 n\rfloor - 1}$ according to Theorem 2.1. So, there is some $s \leq \lfloor 2\log_2 n\rfloor - 1$ such that L represents 4^s primitively and hence we may assume that

$$Q(\mathbf{x}_1) = 4^s,\ B(\mathbf{x}_1, \mathbf{x}_2) \equiv 1 \pmod 2 \ \text{ and } \ B(\mathbf{x}_1, \mathbf{x}_i) \equiv 0 \pmod 2$$

for $3 \leq i \leq n$. If $s \geq 1$, then

$$K := \mathbb{Z}(1/2)\mathbf{x}_1 + \mathbb{Z}2\mathbf{x}_2 + \mathbb{Z}\mathbf{x}_3 + \cdots + \mathbb{Z}\mathbf{x}_n \in \mathfrak{U}_n$$

is adjacent to L and K represents 4^{s-1}. By repeating this procedure, if necessary, we may conclude that there exists $L' \in \mathfrak{U}_{n-1}$ such that

$$d(L, \langle 1\rangle \perp L') \leq s \leq \lfloor 2\log_2 n\rfloor - 1.$$

Therefore

$$\mathfrak{d}(I_n) \leq \lfloor 2\log_2 n\rfloor - 1 + \mathfrak{d}(I_{n-1}).$$

The lemma follows from this and (9). $\square$

We let for $n \geq 17$,

$$\tau_n := \sum_{i=17}^{n} \lfloor 2\log_2 i\rfloor - n + 18 = O(n\log n). \tag{10}$$

From Lemma 3.2 follows that

$$\mathfrak{d}(n) \leq 2\tau_n \ \text{ for } \ n \geq 17.$$

Corollary 3.1 *Let ℓ be any $\mathbb{Z}$-lattice of rank n, $n \geq 14$, with $\mathfrak{s}(\ell) \subseteq 4^{\tau_{n+3}}\mathbb{Z}$. Then $\ell \to I_{n+3}$.*

Proof. Note that $(1/2)^{\tau_{n+3}}\ell \to gen(I_{n+3})$ (see [O'M2]). So there exists $L \in gen(I_{n+3})$ such that

$$(1/2)^{\tau_{n+3}}\ell \to L.$$

Since $2^{\tau_{n+3}}L \to I_{n+3}$ by Lemma 3.2, we have $\ell \to 2^{\tau_{n+3}}L \to I_{n+3}$. $\square$

Theorem 3.2

$$\frac{1}{4\log 2} \leq \lim_{n\to\infty} inf \frac{\mathfrak{d}(I_n)}{n\log n} \leq \lim_{n\to\infty} sup \frac{\mathfrak{d}(I_n)}{n\log n} \leq \frac{2}{\log 2}\ .$$

Proof. The inequality in the middle is trivial. The last inequality is the direct consequence of Lemma 3.2 and the Stirling's formula. Now we prove the first inequality. Since $\Delta(L) < 2^{n+1}$ for all $L \in \mathfrak{U}_n$ by Lemma 3.1,

$$|\mathfrak{U}_n/\simeq| \leq \sum_{i=1}^{\mathfrak{d}(I_n)} 2^{(n+1)i}.$$

Let $M(I_n)$ be the *mass* of the genus of I_n. Then

$$2M(I_n) \leq |\mathfrak{U}_n/\simeq| \leq \frac{2^{(n+1)\mathfrak{d}(I_n)}-1}{2^{n+1}-1}\,.$$

Since

$$M(I_n) \sim \frac{C\left(\dfrac{n}{2\pi e\sqrt{e}}\right)^{n^2/4}\left(\dfrac{8\pi e}{n}\right)^{n/4}}{n^{1/24}}\,,$$

where C is a constant (see [MH]), the first inequality follows. □

Remark. Observe that from the proof of the above theorem, one can easily deduce that

$$h(I_n) = n^{\Theta(n^2)}.$$

4 Upper and lower bounds for $g[n]$

Let $\ell = \mathbb{Z}\mathbf{x}_1 + \mathbb{Z}\mathbf{x}_2 + \cdots + \mathbb{Z}\mathbf{x}_n \subset I_g$ be a $\mathbb{Z}$-lattice, where $\mathbf{x}_i = (a_{i1}, a_{i2}, \cdots, a_{ig})$ for $1 \leq i \leq n$. For $1 \leq j \leq g$, define

$$\mathbf{v}_j := {}^t(a_{1j}, a_{2j}, \cdots, a_{nj}),$$

where ${}^tA$ is the transpose of a matrix or a vector A. Since these $\mathbf{v}_j$'s also characterize ℓ, we define

$$S_n(\mathbf{v}_1, \mathbf{v}_2, \cdots \mathbf{v}_g) := \ell \subset I_g. \tag{11}$$

Note that for positive integers $s_1, s_2, \cdots, s_g$,

$$S_n(\overbrace{\mathbf{v}_1, \cdots \mathbf{v}_1}^{s_1 \text{ copies}}, \overbrace{\mathbf{v}_2, \cdots \mathbf{v}_2}^{s_2 \text{ copies}}, \cdots, \overbrace{\mathbf{v}_g, \cdots \mathbf{v}_g}^{s_g \text{ copies}}) \subset I_{s_1+s_2+\cdots+s_g}$$

is represented by the $\mathbb{Z}$-lattice $\langle s_1 \rangle \perp \langle s_2 \rangle \perp \cdots \perp \langle s_g \rangle$ of rank g.

We define

$$\lambda(n) := \min\{\, g \mid \ell \to I_g \perp \langle 2, \cdots, 2 \rangle \text{ for all } \ell \in \mathfrak{S}_n \,\}.$$

For a sublattice ℓ of $L \perp M$ of the form

$$\ell = \mathbb{Z}(\mathbf{x}_1 + \mathbf{y}_1) + \mathbb{Z}(\mathbf{x}_2 + \mathbf{y}_2) + \cdots + \mathbb{Z}(\mathbf{x}_n + \mathbf{y}_n)$$

with $\mathbf{x}_i \in L$ and $\mathbf{y}_i \in M$, we define

$$\ell(L) := \mathbb{Z}\mathbf{x}_1 + \mathbb{Z}\mathbf{x}_2 + \cdots + \mathbb{Z}\mathbf{x}_n \ \text{ and } \ \ell(M) := \mathbb{Z}\mathbf{y}_1 + \mathbb{Z}\mathbf{y}_2 + \cdots + \mathbb{Z}\mathbf{y}_n.$$

Lemma 4.1

$$\lambda(n) \leq 2^n - 1.$$

Proof. Let $\ell = S_n(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_g) \subset I_g$ be any $\mathbb{Z}$-sublattice of rank n. Assume that $g \geq 2^n$. Then either (i) there exists i satisfying $\mathbf{v}_i \equiv \mathbf{0} \pmod 2$ or (ii) there exist i, j, $i \neq j$ satisfying $\mathbf{v}_i \equiv \mathbf{v}_j \pmod 2$. If we let

$$\ell_0 = S_n\left(\cdots, \mathbf{v}_i/2, \mathbf{v}_i/2, \mathbf{v}_i/2, \mathbf{v}_i/2, \cdots\right)$$

for the case (i), then

$$\ell \cong \ell_0 \to I_{g-1} \perp \langle 4 \rangle \to I_{g-1} \perp \langle 2, 2 \rangle.$$

Consider the case (ii). If $\mathbf{v}_i = \pm\mathbf{v}_j$, then

$$\ell \to I_{g-1} \perp \langle 2 \rangle.$$

If $\mathbf{v}_i \neq \pm\mathbf{v}_j$, let

$$\ell_1 = S_n\left(\cdots, (\mathbf{v}_i + \mathbf{v}_j)/2, (\mathbf{v}_i + \mathbf{v}_j)/2, \cdots, (\mathbf{v}_i - \mathbf{v}_j)/2, (\mathbf{v}_i - \mathbf{v}_j)/2, \cdots\right).$$

Then

$$\ell \cong \ell_1 \to I_{g-2} \perp \langle 2, 2 \rangle.$$

The lemma follows. $\square$

Theorem 4.1 *For all $n \geq 14$,*

$$g[n] \leq 2\lambda(n)\tau_{n+3} + n + 4.$$

Proof. Assume $\ell \to I_g$ with $g > \lambda(n)$. Then $\exists\, \lambda_1, g_1$ such that

$$\ell \to I_{\lambda_1} \perp \sqrt{2} I_{g_1},$$

where $\lambda_1 \leq \lambda(n)$. If $g_1 > \lambda(n)$, Then $\exists\, \lambda_2, g_2$ such that

$$\ell \to I_{\lambda_1} \perp \sqrt{2} I_{\lambda_2} \perp 2 I_{g_2},$$

where $\lambda_2 \leq \lambda(n)$. Repeating this process, if applicable, we obtain

$$\ell \to I_{\lambda_1} \perp \sqrt{2} I_{\lambda_2} \perp \cdots \perp (\sqrt{2})^{t-1} I_{\lambda_t} \perp (\sqrt{2})^t I_{g_t},$$

where $\lambda_j \leq \lambda(n)$ for all $1 \leq j \leq t$ and $t = 2\tau_{n+3}$. If we denote this representation by σ, then $\sigma(\ell)((\sqrt{2})^t I_{g_t})$ is represented by I_{n+3} by Corollary 3.1. Observe that

$$I_{\lambda_1} \perp (\sqrt{2})^2 I_{\lambda_3} \perp \cdots \perp (\sqrt{2})^{t-2} I_{\lambda_{t-1}} \to I_{\lambda_1+\lambda_3+\cdots+\lambda_{t-1}},$$

and

$$\sqrt{2} I_{\lambda_2} \perp (\sqrt{2})^3 I_{\lambda_4} \perp \cdots \perp (\sqrt{2})^{t-1} I_{\lambda_t} \to I_{\lambda_2+\lambda_4+\cdots+\lambda_t+1}.$$

Combining these, we obtain

$$\ell \to I_{\lambda_1+\cdots\lambda_t+n+4}.$$

This proves the theorem. $\square$

Note that Lemma 4.1 and the above theorem together with (10) imply that

$$g[n] = O(2^n\, n \log n). \tag{12}$$

We now improve this upper bound. To do this, we have to improve the upper bound of $\lambda(n)$ of Lemma 4.1. Let $\ell \in \mathfrak{S}_n$. We call ℓ a *terminal* $\mathbb{Z}$-lattice if $\sigma(\ell)(\langle 2, \cdots, 2\rangle) = 0$ for any representation $\sigma : \ell \to I_g \perp \langle 2, \cdots, 2\rangle$. Let $\mathfrak{T}_n$ be the set of all terminal $\mathbb{Z}$-lattices of rank n, which is a subset of $\mathfrak{S}_n$.

Lemma 4.2 *For all $\ell \in \mathfrak{S}_n$, there exist an $\ell' \in \mathfrak{T}_n$ and a representation $\phi : \ell \to I_g \perp \langle 2, \cdots, 2\rangle$ such that*

$$\phi(\ell)(I_g) = \ell'.$$

Proof. Let $\mu_i(\ell)$ be the i-th successive minimum of ℓ. We will use an induction on

$$m(\ell) := \prod_{i=1}^{n} \mu_i(\ell).$$

If $m(\ell) = 1$, then clearly $\ell = I_n$ and hence ℓ itself is a terminal $\mathbb{Z}$-lattice. If $\phi : \ell \to I_g \perp \langle 2, \cdots, 2\rangle$ and $\phi(\ell)(\langle 2, \cdots, 2\rangle) \neq 0$, then $m(\phi(\ell)(I_g)) < m(\ell)$. Hence by induction hypothesis, there exists

$$\phi' : \phi(\ell)(I_g) \to I_h \perp \langle 2, \cdots, 2\rangle,$$

such that $\phi'(\phi(\ell)(I_g))(I_h)$ is a terminal $\mathbb{Z}$-lattice. The lemma follows immediately. $\square$

Remark. From above lemma, we have

$$\lambda(n) = \min\{\, g \mid \ell \to I_g, \text{ for all } \ell \in \mathfrak{T}_n\}.$$

Lemma 4.3

$$\lambda(n)(\lambda(n) - 1) \leq 2 \times 3^n - 1.$$

Proof. For $\ell \in \mathfrak{T}_n$, let $\ell = S_n(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_g)$ and $\mathbf{v}_j \neq \mathbf{0}$ for all $1 \leq j \leq g$. Note that $\mathbf{v}_i \neq \mathbf{v}_j$ for any $i \neq j$ since $\ell \notin \mathfrak{T}_n$ otherwise. We may assume that g is the minimum rank in order for I_g to represent ℓ. Suppose that $g(g-1) \geq 2 \times 3^n$. Then, by the pigeon hole principle, either (i) there exist two distinct indices i, j such that $\mathbf{v}_i \equiv \mathbf{v}_j \pmod 3$, or (ii) there exist four distinct indices i, j, k, l such that

$$\mathbf{v}_i + \mathbf{v}_j \equiv \mathbf{v}_k + \mathbf{v}_l \pmod 3.$$

For the case (i), note that

$$S_n(\mathbf{v}_i, \mathbf{v}_j) \simeq S_n\Big(\overbrace{(\mathbf{v}_i - \mathbf{v}_j)/3, \cdots, (\mathbf{v}_i - \mathbf{v}_j)/3}^{\text{4 copies}}, (\mathbf{v}_i + 2\mathbf{v}_j)/3, (-2\mathbf{v}_i - \mathbf{v}_j)/3\Big).$$

This contradicts to the fact that $\ell \in \mathfrak{T}_n$. Consider the case (ii). If $\mathbf{v}_i + \mathbf{v}_j = \mathbf{v}_k + \mathbf{v}_l$, then

$$S_n(\mathbf{v}_i, \mathbf{v}_j, \mathbf{v}_k, \mathbf{v}_l) \to A_3 \to I_3,$$

which is not possible. So, $\mathbf{v}_i + \mathbf{v}_j \neq \mathbf{v}_k + \mathbf{v}_l$. One can easily show that

$$S_n(\mathbf{v}_i, \mathbf{v}_j, \mathbf{v}_k, \mathbf{v}_l) \simeq S_n\left(\mathbf{w}, \mathbf{w}, \mathbf{w} + \mathbf{v}_l, \mathbf{w} + \mathbf{v}_k, \mathbf{w} - \mathbf{v}_j, \mathbf{w} - \mathbf{v}_i\right),$$

where $\mathbf{w} = (\mathbf{v}_i + \mathbf{v}_j - \mathbf{v}_k - \mathbf{v}_l)/3$. This again contradicts to the fact that $\ell \in \mathfrak{T}_n$. $\square$

Corollary 4.2 *For $n \geq 14$,*

$$g[n] \leq 3 \cdot 3^{n/2}\, \tau_{n+3} + n + 4.$$

In particular,

$$g[n] = O(3^{n/2}\, n \log n).$$

Proof. This is an immediate consequence of (10), Theorem 4.1 and Lemma 4.3. $\square$

Remark. As we can see from Lemmas 4.1 and 4.3, it seems that this bound for $\lambda(n)$ can be improved further. But it is doubtful that our method may give us a polynomial bound for $\lambda(n)$ and hence for $g[n]$.

We now provide a lower bound of $g[n]$ in the following theorem.

Theorem 4.3

$$g[n] \geq 2n - \lceil \log_2 n \rceil - 2.$$

Proof. Let m, n be positive integers satisfying $n \leq 2^m$. It suffices to show

$$g[n + m + 1] \geq 2n + m. \tag{13}$$

If we let $N = \{1, 2, \cdots, n\}$, then we can always find m subsets $T_1, T_2, \cdots, T_m$ of N such that for any $\{i, j\} \subset N$, there exists an index k for which

$$|T_k \cap \{i, j\}| = 1.$$

Let $\{e_1, \cdots, e_n, e_{n+1}, \cdots, e_{n+m}, f_1, \cdots, f_n\}$ be an orthonormal basis of I_{2n+m}. If we define

$$\ell := \sum_{i=1}^{n} \mathbb{Z}(e_i + f_i) + \sum_{k=1}^{m} \left(\mathbb{Z} \left(\sum_{j \in T_k} e_j + e_{n+k} \right) \right) + \mathbb{Z}(f_1 + \cdots + f_n),$$

then one can easily show from the definition of T_k that

$$\ell \to I_{2n+m} \quad \text{but} \quad \ell \not\to I_{2n+m-1}.$$

From this follows (13) and the theorem. $\square$

Remark. For small n's, say $7 \le n \le 20$, much sharper bounds are known (see [KO3]). For example, $11 \le g[7] \le 24$, $13 \le g[8] \le 37$, and

$$g[n] \le \begin{cases} n(n+1)/2 + n + 3 & \text{for } 9 \le n \le 11, \\ 5n^3/2 & \text{for } 12 \le n \le 20. \end{cases} \tag{14}$$

5 s-integrable $\mathbb{Z}$-lattices

We apply our result to s-integrable $\mathbb{Z}$-lattices introduced by Conway and Sloane [CS3]. A $\mathbb{Z}$-lattice ℓ is said to be *s-integrable* if $\sqrt{s}\ell$ can be represented by a sum of squares, where s is a positive integer. Define

$$\phi(s) := \min\{\, n \mid \exists \text{ a } \mathbb{Z}\text{-lattice } \ell \text{ of rank } n \text{ such that } \sqrt{s}\ell \notin \mathfrak{S}_n \,\}.$$

It is known [CS3] that

$$\phi(1) = 6, \ \phi(2) = 12, \ \phi(3) = 14, \ \cdots$$

and that

$$\phi(s) \ge 2 \left(\frac{\ln \ln s}{\ln \ln \ln s} (1 + o(1)) \right).$$

We improve this bound in the following theorem.

Theorem 5.1 *Let $q(n) = (\lfloor n^2/4 \rfloor + 1)$ for a positive integer n. Then for any integer s satisfying*

$$\left((q(n)+1)^{2n} - 1 \right) q(n)^{2n} \le s \le \left((q(n+1)+1)^{2(n+1)} - 1 \right) q(n+1)^{2(n+1)},$$

$\phi(s) \geq n-2$. *In particular, for large* s,

$$\phi(s) \geq \frac{\ln s}{8 \ln \ln s}.$$

Proof. Let $L = \mathbb{Z}\mathbf{x}_1 + \mathbb{Z}\mathbf{x}_2 + \cdots + \mathbb{Z}\mathbf{x}_n$ be an odd unimodular $\mathbb{Z}$-lattice of rank n. Then by Theorem 2.1, L represents q^2, where $q := q(n)$. So, we can find a divisor b of q such that b^2 is primitively represented by L and hence by a suitable base change, if necessary, we may assume that

$$Q(\mathbf{x}_1) = b^2,\ \gcd(B(\mathbf{x}_1, \mathbf{x}_2), b) = 1,\ B(\mathbf{x}_1, \mathbf{x}_i) \equiv 0 \pmod{b}$$

for $3 \leq i \leq n$. Furthermore we may assume that

$$bQ(\mathbf{x}_2) \not\equiv B(\mathbf{x}_1, \mathbf{x}_2) \pmod{2}.$$

Let

$$\tilde{L} := \mathbb{Z}(1/b)\mathbf{x}_1 + \mathbb{Z}b\mathbf{x}_2 + \mathbb{Z}\mathbf{x}_3 + \cdots + \mathbb{Z}\mathbf{x}_n.$$

Then $\tilde{L} \cong \langle 1 \rangle \perp L'$, where L' is an odd unimodular $\mathbb{Z}$-lattice of rank $n-1$. So

$$qL \to \langle 1 \rangle \perp L'.$$

Since L' also represents q^2 by Theorem 2.1, we may conclude that

$$q^n L \to I_n \text{ and similarly } (q+1)^n \ell \to I_n.$$

Thus for every integer $s \geq ((q+1)^{2n} - 1)q^{2n}$, L is s-integrable. Since every $\mathbb{Z}$-lattice ℓ of rank $n-3$ is represented by $gen(I_n)$, ℓ is s-integrable. The theorem follows. $\square$

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